

Project Executable Files

Date	29 Jun 2026
Team ID	xxxxxx
Project Name	OptiCrop
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL	Yes
7	APK / Executable binary	NA
8	Dockerfile / Containerization config	NA
9	Test files and test results	Yes
10	Demo video or walkthrough	No

Step 2: File / Folder Structure

OptiCrop/

static/	- CSS, JavaScript, Images
templates/	- HTML templates (Frontend pages)
venv/	- Python virtual environment (not uploaded to GitHub)
app.py	- Main Flask application
train_model.py	- Machine learning model training script
crop_model.pkl	- Trained machine learning model
Crop_recommendation.csv	- Crop recommendation dataset
database.py	- Database-related operations
requirements.txt	- Project dependencies
README.md	- Project setup and execution guide
.gitignore	- Git ignore configuration

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-m04h.onrender.com
Login Credentials (Demo)	Mail: daganisuryaprakash@gmail.com Password: Surya@1234 pass
Platform / Hosting Provider	
Repository Link	https://github.com/rockychandu/SmartbridgeProject
Demo Video Link	

Step 4: Run Instructions

Pre-requisites

- Python 3.10 or above
- Flask
- NumPy
- Pandas
- Scikit-learn
- Joblib
- Web browser (Chrome, Edge, or Firefox)
- Git (optional, for cloning the repository)

Step 1

Clone the repository:

```
git clone <GitHub Repository URL>
```

Step 2

Navigate to the project directory:

```
cd OptiCrop
```

Step 3

Install the required dependencies:

```
pip install -r requirements.txt
```

Step 4

Run the Flask application:

```
python app.py
```

Step 5

Open the URL displayed in the terminal (typically `http://127.0.0.1:5000/`) in your web browser.

Step 6

Enter the required soil and environmental parameters:

- Nitrogen (N)
- Phosphorous (P)

- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

Click **Predict** to view the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The application runs only on a local Flask server if not deployed.	Can be deployed on Render or another cloud platform.
2	Prediction accuracy depends on the quality and size of the training dataset.	Use a larger and more diverse dataset to improve accuracy.
3	No user authentication or account management is implemented.	Can be added as a future enhancement if required.